SWRNet: A Deep Learning Approach for Small Surface Water Area Recognition Onboard Satellite

Trong-An Bui, *Member, IEEE*, and Pei-Jun Lee, *Senior Member, IEEE*

***Abstract*—This article proposes a deep learning approach for small surface water recognition using multispectral satellite imaging, which reduces the computational complexity by 18.66 times and increases the accuracy of surface water recognition by up to 14.1%. The proposed model uses near infrared combined with RGB spectral imagery to increase the accuracy of surface water recognition. In addition, since surface water only accounts for a small percentage of the remote sensing dataset, thus creating an imbalance problem, a proposed loss function is introduced to combine region-based and distribution-based loss. This article introduces an adaptive factor that automatically adjusts the weighting between distribution- and region-based loss functions. The proposed adaptive factor is determined based on the loss value of the previous training step. The mean intersection over union of surface water between predicted and ground truth regions is recorded as 0.80.**

***Index Terms*—Compound loss function, cutting-edge model, remote sensing, small surface water recognition.**

I. Introduction

Remote sensing provides a lower cost and much more frequent revisit cycles, which is currently adopted in environmental monitoring [1], flood assessment [2], and hydroelectric power generation [3] due to its precise recognition of surface water. This study introduces a deep learning model designed to identify water areas on Earth's surface.

In remote sensing object recognition, works in [4], [5], and [6] use DeepLab [7], [8]. Fully convolutional networks (FCNs) [9] are used in [10], [11], and [12]. U-NET [13] is selected as a deep learning model in [14] and [15]. Deep learning models (FCN, U-NET, DeepLab) have higher accuracy than traditional/nondeep learning methods (watershed [16] or Otsu [22]) [17], but the computation complexity is much higher. In these architectures, satellite images are encoded and transmitted to the ground station for decoding. Then, a deep learning model is used to identify the object. Thus, the identification and classification of the target objects have a considerable latency compared to real-time. Thus, the aforementioned architectures are efficient for urban planning studies or using satellite imagery for long-term purposes.

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The authors are with the Electronic and Computer Engineering Department, National Taiwan University of Science and Technology, Taipei 106335, Taiwan (e-mail: trongan93@mail.ntust.edu.tw; pjlee@mail.ntust.edu.tw).

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This article proposes a high-accuracy satellite payload deep learning model for real-time surface water recognition. The proposed deep learning is a lightweight model with nanosatellite specifications, such as IRIS-B, a Cube 3U satellite (CubeSAT) [19], significantly reducing signal transmission time and power consumption. To download a 1920 × 1200 RGB image from satellites to ground station requires a minimum transmission of 6.912 MB. To accurately monitor surface water areas, the satellite camera sensor needs to record at 5–12 frames per second, which leads minimum transmission required of 34.56 to 82.944 MB per second. Because CubeSat download bandwidth is limited, images captured in real-time in the satellite payload are delayed transfer to the ground station. Conventional remote sensing methods at ground stations are impossible to make timely surface water identifications in real-time. Increasing the download capacity of the satellite to the ground station is one solution, but this method consumes a large amount of battery power on the small satellite, exceeds the limited power of CubeSAT, and increases the cost of the ground station construction.

This study introduces a model that can respond to real-time identification. In the proposed model, the surface water recognition is calculated onboard the satellite and is sent straight to the ground station as much smaller data products that take significantly less time to transmit. The surface water map is only 0.288 MB, which is 24 times smaller than transmitting an RGB image for a 1920 × 1200 camera sensor. The applications based on surface water map are available much earlier than when calculated at the ground station.

Onboard recognition model strikes a balance between accuracy and computational complexity, effectively reconciling speed with the accuracy of area identification, particularly in surface water regions. In remote sensing object recognition, deep learning models are mainly built based on FCNs [18], U-NET [13], and DeepLab [19]. The recognition accuracy of the U-NET model is higher than that of an FCN but lower than DeepLab [17]. In contrast, the U-NET's computation complexity and weighting (parameter size) are greater than FCN and smaller than DeepLab. Essential extensions that involve UNET combined with attention module and transformers, including Attention-Unet [20], [21], TransUNet [22], Swin-Transformer-UNet [23], [24], and Vision-transform-UNet [25], further enhance segmentation performance within remote sensing. However, due to their inclination to incorporate more intricate modules, the time needed for identification onboard satellites experiences delays. In contrast, lightweight models, MALUNet [26] and T-NET [27] boasting reduced computational complexity, compromise identification accuracy, especially concerning small objects, such as water bodies in satellite images. Given the central focus of this research on cutting-edge deep learning design, it becomes imperative to strike a balance between accuracy and computational complexity. This article proposes a deep learning approach for Small surface Water area Recognition (SWRNet), meticulously tailored to adhere to onboard nanosatellite specifications. This approach ensures swift water body recognition without sacrificing accuracy.

Small-scale aquatic features, representing bodies of water with comparatively reduced dimensions in contrast to vast counterparts, such as oceans, seas, and large lakes, comprise a diverse array, including ponds, streams, creeks, rivers, rivulets, wetlands, lakes, reservoirs, lagoons, tarns, and pools. The discernment and systematic monitoring of these diminutive water bodies are indispensable for ecological preservation, efficient water resource management, and the comprehensive examination of climate change implications. Moreover, the ability to anticipate alterations in these small water bodies holds promise for predicting natural disasters, specifically floods, or projecting lake flow dynamics to prevent dam failure, thereby enabling proactive measures and enhancing community resilience. In the realm of remote sensing, the classification of water bodies as small is influenced not only by their reduced scale but also by the impact of spatial resolution. High-spatial-resolution satellite images diminish the apparent size of water bodies. Consequently, this article focuses on identifying small water bodies, emphasizing the importance of precise recognition within the context of remote sensing research. The world flood [2] is an imbalanced dataset in which the water pixels (*blue area*) account for only 3% of the pixels, much lower than other classes—land (43%, *brown area*) and cloud (50%, *gray area*) pixels. This leads to low accuracy in identifying areas of water. Data augmentation techniques based on GAN [28] tackle the imbalanced dataset by creating synthetic water areas. However, this technique only focuses on training set problems.

Because surface water only accounts for a small percentage of a satellite image, in order to increase the identification accuracy in remote sensing, small-object-mining-based network optimization [29] is used in the inference process, which introduces dual-branch (foreground and background) decoding and a collaborative probability loss function. An end-to-end self-cascaded network [30] improves the labeling coherence with sequential global-to-local context aggregation in high-resolution satellite images. A deep learning approach [31] combines segmentation and the edge detection network operating on encoded features to improve the ratio imbalance between foreground and background in high-resolution satellite images. However, these methods increase the number of layers and extra feature calculations on each layer, which increases the identification accuracy but reduces the recognition speed.

This research proposes a deep learning approach for small surface water recognition onboard satellite images. The rest of this article is organized as follows. Section II proposes SWRNet: a deep learning approach for small surface water area recognition onboard satellite, in which a deep learning model with multispectral [combined near infrared (NIR) and RGB] imagery enhances the indistinct areas of optical imagery, which is suitable for nanosatellite payload missions (CubeSAT), and proposes loss function resolves the imbalanced dataset inherent to satellite imagery of areas with little surface water. Section III discusses experiment results. Following the main contributions in the proposed section, Section III first compares the recognition accuracy, focusing on areas of water, and the computation complexity of the proposed cutting-edge deep learning model and other models. Then demonstrates the proposed loss function's effectiveness through the surface water recognition accuracy. Finally, flood prediction is determined based on global river data and recognized surface water by proposed deep learning approach. Section IV presents the limitations of the proposed method. Finally, Section V concludes this article.

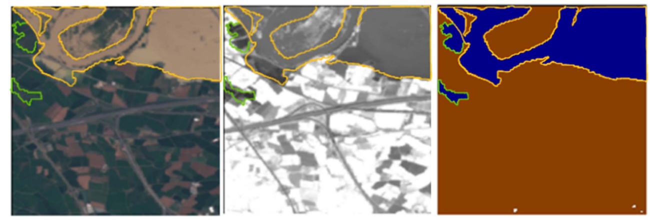


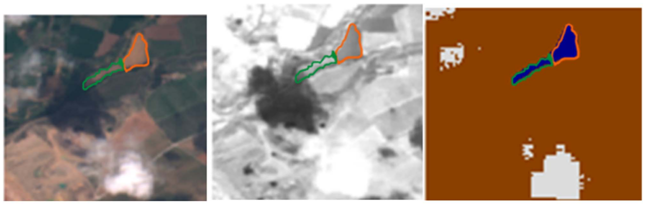
Fig. 1. IR enhances surface water areas on satellite images. (a) RGB image. (b) NIR image. (c) Ground truth of water (blue), land (brown), and cloud (gray) areas.

II. Proposed Cutting-Edge Deep Learning Model for Surface Water Recognition on Satellite Images

In remote sensing, optical satellite images (RGB—combined of Sentinel-2 [32] multispectral bands 4, 3, and 2) are affected different lighting conditions, making them difficult to interpret. Water absorbs more energy (low reflectance) at NIR wavelengths, whereas nonwater reflects more energy (high reflectance) [33], this difference in reflectance values allows us to distinguish water and nonwater areas. Fig. 1(a), (b), and (c) is the RGB (combined of Sentinel-2 multispectral bands 4, 3, and 2), NIR (band 8), and ground truth pixel–map (from World Floods [2]) of a satellite image, respectively.

In Fig. 1, the *yellow* line encircles regions of easily clustered water, clearly recognizable in RGB images. Meanwhile, the green line indicates areas that are not easily distinguishable in the RGB image but can be distinctly identified in the NIR image. It is important to note that NIR observations are indispensable for gaining insights into the material quality of the objects [34].

In Fig. 2, the *orange* and *green* lines indicate a body of water. In the NIR image [see Fig. 2(b)], this area of water is divided into two different regions but the pixel values in the RGB image [see Fig. 2(a)] represent the same area of water to be identified. Thus, in Fig. 2, the area of water is more accurately represented in the RGB image than in the NIR image. By using a combination of both RGB and NIR sensory imaging, we are able to more accurately identify areas of water in a wider range of ground/weather conditions than by using either spectral imagery alone [35]. This article proposes a deep learning model with



(a) (b) (c)

Fig. 2. RGB enhances surface water areas on satellite images. (a) RGB image. (b) NIR image. (c) Ground truth of water (blue), land (brown), and cloud (gray) areas.

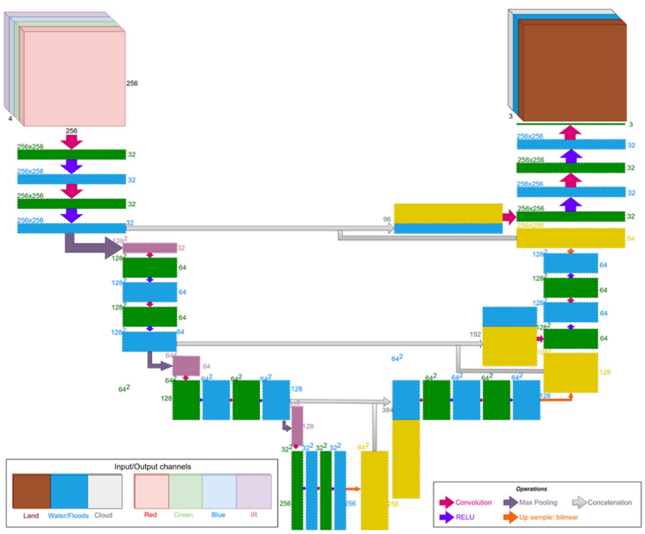


Fig. 3. SWRNet: Proposed cutting-edge deep learning model focused on surface water recognition.

multispectral (R, G, B, and NIR) imagery to improve surface water recognition on satellite images, as illustrated in Fig. 3.

The proposed network inputs a tensor with four image channels (R, G, B, and NIR) and yields a labeled map with land, cloud, and water areas, denoted as $P_{(x,y)}$. The ground truth segmentation is denoted as $Y_{(x,y)}$, which is the pixel-map of land, cloud, and water regions in a satellite image. Each vertical encoder layer uses a double convolution (two-pair Convolution and ReLu) and Max pooling operators. Meanwhile, the vertical decoder layers include concatenation, double convolution, and bilinear interpolation up-sampling layers.

In order to build a complete application of Edge-AI on a satellite payload, this study conducts deep learning model training at the ground station. The parameter, the neural network's output, is then updated and uploaded to the satellite. Because of the upload bandwidth limitation, the parameter size is kept as light as possible. Table I presents the number of each convolution layer and compares the proposed SWRNet with other U-NET-based models.

The computation complexity and parameter size of SWRNet are 10.69 GMac and 1.95 M, respectively, rendering it suitable for power-constrained scenarios and the upload link capacity of a CubeSAT mission [36]. Nevertheless, the utilization of a

TABLE I
KERNEL COMPARISON OF THE PROPOSED MODEL AND OTHER U-NET MODELS

		U-NET [13]	Simple U-NET [2]	Proposed SWRNet
Vertical-layers ($\mathcal{L}$)		4	3	3
Horizontal-layer 1 ($\mathcal{L}_1$)	Convolution	260	260	132
	RELU	256	256	128
	Pooling	64	64	32
	Up-sample	64	64	32
$\mathcal{L}_2$	Convolution	512	512	256
	RELU	512	512	256
	Pooling	128	128	64
	Up-sample	128	128	64
$\mathcal{L}_3$	Convolution	1024	1024	512
	RELU	1024	1024	512
	Pooling	256	256	128
	Up-sample	256	256	128
$\mathcal{L}_4$	Convolution	2048	1024	512
	RELU	2048	1024	512
	Pooling	512	-	-
	Up-sample	512	-	-
$\mathcal{L}_5$	Convolution	2048	-	-
	RELU	2048	-	-

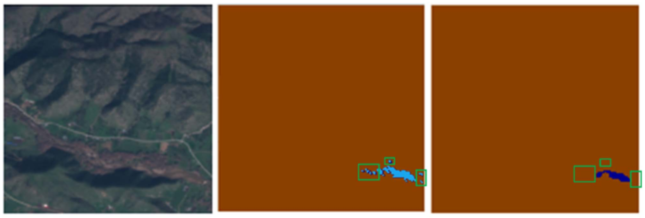
small number of kernels by SWRNet for accelerated recognition adversely affects the accuracy of water body recognition. This article introduces a novel loss function with a focus on enhancing water recognition accuracy in satellite images.

Areas of water are tiny and represent a very small proportion of the dataset compared to others (land and cloud) [2]. The total number of pixels representing water accounts for only 3%, compared with 43% for land and 50% for clouds, the distribution of examples across the classes is imbalanced. The determination of loss values in image recognition in general, and U-NET, in particular, complies with the pixelwise loss function [37]. The contribution of land and cloud classes, which are more frequently represented in the dataset, to the overall loss calculation and weight updates during the backward model training process is comparatively larger. This is because the loss values and weight updates are influenced by the frequency and distribution of these classes, whereas the contribution of the water class, which has a smaller representation in the dataset, is relatively lower in the loss function and weight updates. Hence, the loss values depend on the classes with larger pixel ratios, namely land, and clouds, due to their higher representation in the dataset.

The weighted cross-entropy ($L_{\mathrm{BWCE}}(y, p_t)$) [38] and focal ($L_{FC}(y, p_t)$) are distribution-based loss functions [39], which minimize dissimilarity between two distributions of unequal samples at a training step (t), which assign more weight to the loss value of the water pixels. This suggests that the model should prioritize the accurate prediction of water pixels to improve the accuracy of water area identification. y and p_t are the pixel values of $Y(x, y)$ and $P_t(x, y)$, respectively. In which, $L_{FC}(y, \widehat{p}_t)$ adopts an automatic adjustment of the weight factor during the training process

$$L_{FC}(y, p_t) = -\alpha(1 - p_t)^{\beta} y \log(p_t) \tag{1}$$

where $p_t = \begin{cases} p_t, & y = 1 \\ 1 - p_t, & \text{otherwise} \end{cases}$ and $\alpha = \begin{cases} \alpha & , y = 1 \\ 1 - \alpha, & \text{otherwise} \end{cases}$.



(a) (b) (c)

Fig. 4. Surface water recognition from satellite images. (a) RGB image. (b) Ground truth $Y_W(x, y)$. (c) Predicted area of water ($P_{W_t}(x, y)$).

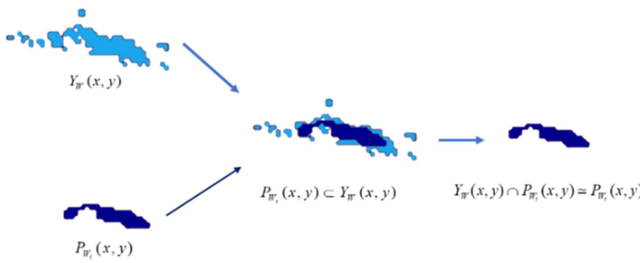


Fig. 5. Region relationship between $P_{W_t}(x, y)$ and $Y_W(x, y)$.

Furthermore, region-based loss function minimizes the mismatched water areas ($Y_W(x, y) \neq P_{W_t}(x, y)$) and maximizes the overlap regions ($Y_W(x, y) \cap P_{W_t}(x, y)$), where $Y_W(x, y)$ and $P_{W_t}(x, y)$ are the water area in ground truth ($Y_{(x,y)}$) and ($P_{(x,y)}$) and prediction map, respectively. A region-based loss function is used to resolve the small region identification in satellite images. Dice and Intersection over Union (IoU) losses are two region-based loss functions based on Dice (2) and Jaccard/IoU (3) coefficients, which are used to define the mismatch and overlap water regions between $Y_W(x, y)$ and $P_{W_t}(x, y)$, respectively

$$D_{(Y_W(x,y),P_{W_t}(x,y))} = \frac{2 \times |Y_W(x, y) \cap P_{W_t}(x, y)|}{|Y_W(x, y)| + |P_{W_t}(x, y)|} \tag{2}$$

$$\mathrm{IoU}_{(Y_W(x,y),P_W(x,y))} = \frac{|Y_W(x, y) \cap P_{W_t}(x, y)|}{|Y_W(x, y) \cup P_{W_t}(x, y)|}. \tag{3}$$

Fig. 4 shows the surface water recognized by the U-NET model [13]. Fig. 4(a), (b), and (c) is the RGB image, ground truth ($Y_W(x, y)$), and predicted areas of water ($P_W(x, y)$), respectively.

In the identification of water bodies [$darkblue$ regions in Fig. 4(c)] of small size, the identified area is usually smaller and a subset of the ground truth [$lightblue$ regions in Fig. 4(b)], which means $P_{W_t}(x, y) \subset Y_W(x, y)$ (see Fig. 5).

The identified area of water is lost and replaced by the background [the blue regions in green squares of Fig. 4(b) and (c)]. In the case $P_{W_t}(x, y) \subset Y_W(x, y)$, the Dice and IoU loss function are defined as (4) and (5), respectively

$$L_D(Y_W(x, y), P_W(x, y)) = 1 - D_{(Y_W(x,y),P_{W_t}(x,y))}$$

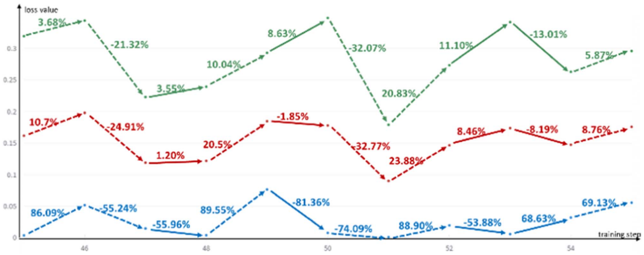


Fig. 6. Rate of change and effect of $L_{FC}(y, p_t)$ and $L_{IoU}(y, p_t)$ against $L_{\text{combo}}(y, p_t)$.

$$= 1 - \frac{2 \times P_{W_t}(x, y)}{Y_W(x, y) + P_{W_t}(x, y)}, \text{where} P_{W_t}(x, y) \subset Y_W(x, y) \tag{4}$$

$$L_{IoU}(Y_W(x, y), P_{W_t}(x, y)) = 1 - IoU_{(Y_W(x,y),P_{W_t}(x,y))}$$
$$= 1 - \frac{P_{W_t}(x, y)}{Y_W(x, y)}, \text{where } P_{W_t}(x, y) \subset Y_W(x, y). \tag{5}$$

In the case $P_{W_t}(x, y) \subset Y_W(x, y)$, $L_{IoU}(Y_W(x, y), P_{W_t}(x, y)) = L_D(Y_W(x, y), P_W(x, y))$, where $p_t = 1$ (absolutely accurate identification) and $L_{IoU}(Y_W(x, y), P_{W_t}(x, y)) < L_D(Y_W(x, y), P_W(x, y))$, where $p_t < 1$. Therefore, this study selects $L_{IoU}(Y_W(x, y), P_{W_t}(x, y))$ as a region-based loss to increase the accuracy of water body recognition.

To solve the imbalanced dataset and identify small objects, studies [2], [40], [41] introduce the combo loss, which averages distributed-based and region-based loss functions on each training step (t). Equation (6) is a combo loss between Focal and IoU

$$\begin{aligned} L_{\text{combo}}(y, p_t) &= \lambda_1 \times L_{FC}(y, p_t) + \lambda_2 \times L_{IoU}(y, p_t) \\ &= \lambda_1 \times (-\alpha(1 - p_t)^{\beta} y \log(p_t)) \\ &\quad + \lambda_2 \times \left(1 - \frac{y \cdot p_t}{y + p_t - y \cdot p_t}\right) \end{aligned} \tag{6}$$

where λ_1 and λ_2 are the factors that control the weighted sum of $L_{FC}(y, p_t)$ and $L_{IoU}(y, p_t)$. Because the combo loss averages the loss value member, $\lambda_1 = \lambda_2 = 0.5$. Fig. 6 presents $L_{\text{combo}}(y, p_t)$ values for each training step. The green, blue, and red lines represent $L_{FC}(y, p_t)$, $L_{IoU}(y, p_t)$, and $L_{\text{combo}}(y, p_t)$, respectively. The dashed and solid lines represent the uniform and nonuniform between $L_{FC}(y, p_t)$ and $L_{IoU}(y, p_t)$, respectively.

In training steps *45*, *46*, *48*, *50*, *51*, and *54*, the variation of $L_{FC}(y, p_t)$ and $L_{IoU}(y, p_t)$ values is uniform (increasing or decreasing) and is represented by dashed lines. Hence, the average loss function—$L_{\text{combo}}(y, p_t)$—has the same volatility as the two membership loss functions. However, in training steps 47, 49, 52, and 53, the volatility of member loss functions is different and is represented by solid lines. When $t = 47, 49, 52$, $L_{FC}(y, p_t)$ fluctuates with increasing value, whereas $L_{IoU}(y, p_t)$ value is decreased. In contrast, when $t = 53$, $L_{FC}(y, p_t)$ value is decreased and $L_{IoU}(y, p_t)$ is increased.

TABLE II
LOSS FUNCTIONS WITH IMBALANCED DATASET AND SMALL/TINY OBJECT IDENTIFICATION

	Imbalanced dataset	Small/Tiny object identification	Adaptive/ Auto adjustment
L_{BWCE}	✓		
L_{FC}	✓		✓
L_D		✓	
L_{IoU}		✓	
$L_{combo1} = \lambda_1 \times L_{BWCE} + \lambda_2 \times L_D$	✓	✓	
$L_{combo2} = \lambda_1 \times L_{FC} + \lambda_2 \times L_D$	✓	✓	✓ (L_{FC} only)
$L_{proposed}$	✓	✓	✓

More specifically, in training step 47, $L_{IoU}(y, p_t)$ is decreased by 55.96% and $L_{FC}(y, p_t)$ is increased by 3.55%. Because the value domain of $L_{FC}(y, p_t)$ and $L_{IoU}(y, p_t)$ is significantly different, the final loss function $L_{\text{combo}}(y, p_t)$, on average member losses, increases by 1.20%. This makes training weighting dependent on $L_{FC}(y, p_t)$ to solve the imbalanced dataset problem but ignores $L_{IoU}(y, p_t)$ to solve the identification of small objects.

This article proposes a new loss function, which solves not only the low distribution label (water distribution), but also small area recognition (the flood prediction area is a subset of the ground truth). Table II compares the objective of L_{proposed} and other loss functions used in object recognition deep learning models, L_{BWCE}, L_{FC}, L_D, L_{IoU}, L_{combo1} [42], [2], L_{combo2} [40]. The proposed loss function is focused on the imbalanced dataset (the low water distribution label), small region identification (tiny areas of water on the satellite image), and adaptive/auto adjustment to each loss member. With the proposed loss function, the neural network on the payload self-adjusts and adapts without remote adjustment via upload bandwidth from the ground station

$$L_{\text{proposed}}(y, p_t) = \frac{1}{-\alpha(1-p_{t-1})^{\beta} y \log(p_{t-1})} \times (-\alpha(1-p_t)^{\beta} y \log(p_t)) + \frac{1}{1 - \frac{y \cdot p_{t-1}}{y + p_{t-1} - y \cdot p_{t-1}}} \times \left(1 - \frac{y \cdot p_t}{y + p_t - y \cdot p_t}\right) \tag{7}$$

where $(t-1)$ is the previous training step.

Equation (7) is the proposed adaptive loss function, which controls the influence of each loss member by the division of 1 with the value of each loss member in the previous training step.

Fig. 7 presents the loss values of each training step in the first epoch of the proposed loss function. The green, blue, and red lines represent $L_{FC}(y, p_t)$, $L_{IoU}(y, p_t)$, and $L_{\text{proposed}}(y, p_t)$, respectively. $L_{IoU}(y, p_t)$ value at training step 43 is increased 12.14 times compared to the previous training step. Meanwhile, $L_{FC}(y, p_t)$ and $L_{\text{proposed}}(y, p_t)$ values are increased by 1.03 and 2.37 times, respectively. In training step 44, $L_{IoU}(y, p_t)$ value decreased by 1.75 times, but focal value increased only by 1.1 times. The value of the proposed compound loss is reduced by 1.81 times compared with the previous training step. This proves that the proposed compound loss function is influenced by both $L_{IoU}(y, p_t)$ and $L_{FC}(y, p_t)$ functions. Therefore, the neural network weight is updated and influenced by both $L_{IoU}(y, p_t)$ and $L_{FC}(y, p_t)$, which means that the water identification accuracy is improved.

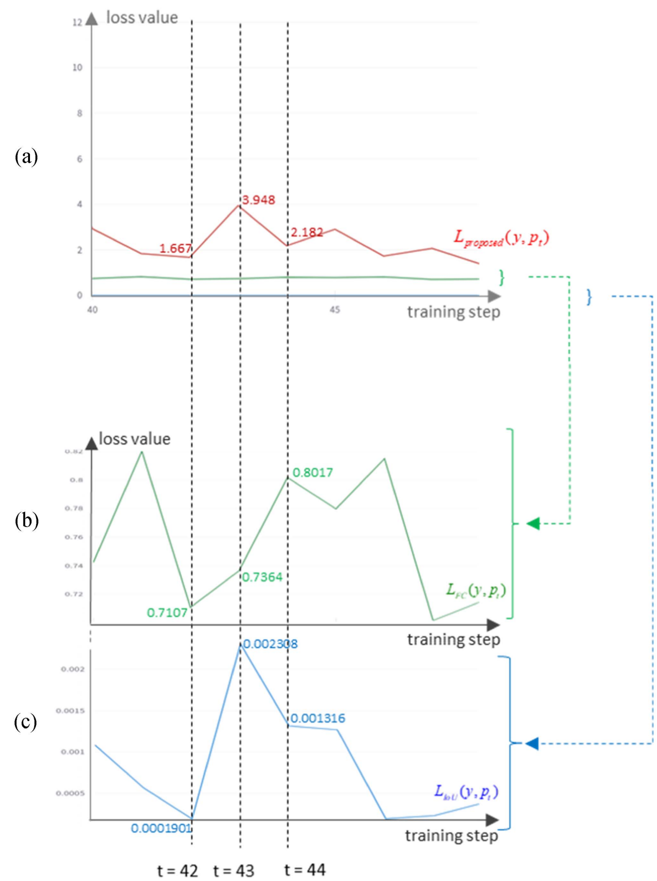


Fig. 7. Comparison of loss values between proposed loss with each loss member. (a) Representation of $L_{FC}(y, p_t)$, $L_{IoU}(y, p_t)$, and $L_{\text{proposed}}(y, p_t)$ in training steps from 40 to 50. (b) Zoom-in scaled values of $L_{FC}(y, p_t)$ from Fig. 8(a). (c) Zoom-in scaled values of $L_{IoU}(y, p_t)$ from Fig. 8(a).

In back-propagation, the derivative of the proposed loss respects a specific neural network weight

$$\frac{\partial L_{\text{proposed}}(y, p_t)}{\partial p_t} = \frac{1}{(1-p_{t-1})^{\beta} y \log(p_{t-1})} \times \left(\frac{y}{p_t}(1-p_t)^{\beta} - \beta(1-p_t)^{\beta-1}\right) + \frac{1}{1 - \frac{y \cdot p_{t-1}}{y + p_{t-1} - y \cdot p_{t-1}}} \times \frac{y^2}{(y + p_t - y p_t)^2}. \tag{8}$$

The proposed SWRNet weighting values [(8) results] are also calculated based on the autoadaptation of derivative $L_{IoU}(y, p_t)$ and $L_{FC}(y, p_t)$.

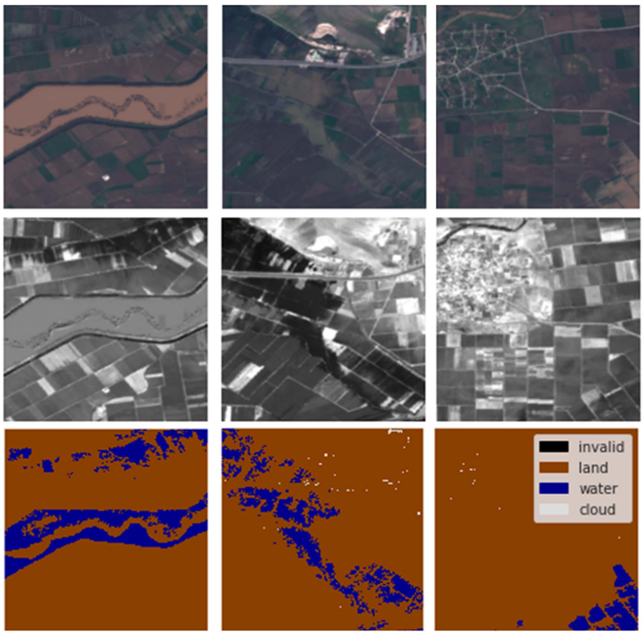


Fig. 8. Example cases in the Floods dataset [2].

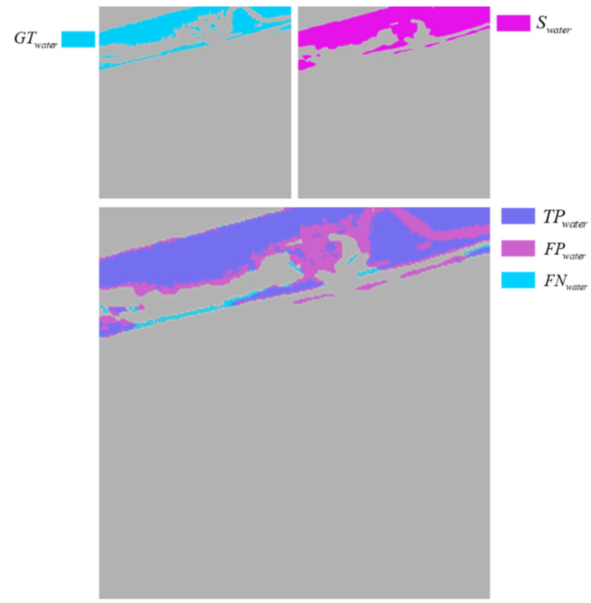


Fig. 9. Intrusion of (10) with water label.

III. Experiment Results

The World Floods database [2] contains 444 pairs of Sentinel-2 satellite images and segmentation masks, which includes 422 water surface maps created by photo interpretation either manually or semiautomatically, where a human validated machine-generated maps. We split the dataset images into 194 151 and 1284 subimages with a size of 256 × 256 to build the training and validation dataset. Fig. 8 shows the example cases in the dataset; the first-row images are RGB, captured and split from Sentinel-2 satellites. The second row shows the NIR images of the same locations as the first row. The third row is the ground truth surface water map.

This section highlights the effectiveness of the proposed deep learning model (SWRNet) for pinpointing small water bodies, highlighting its high accuracy and low computational complexity, making it suitable for onboard satellite deployment. Its applications extend to wetlands detection, environmental monitoring [1], flood assessment [2], water reservoir management, and hydroelectric power generation [3]. In addition, the section validates the efficacy and importance of the study through a flood prediction algorithm based on the proposed model's small water area determinations.

A. Surface Water Area Recognition

This section presents the results of surface water area recognition using the proposed SWRNet model with satellite images.

In order to evaluate the effectiveness of identifying surface areas in satellite images, more specifically, to assess the accuracy of identifying water bodies, mean IoU ($mIoU_{\text{label}}$) and mean Average Precision (mAP_{label}) are confusion matrices comparing image segmentation accuracy. Equation (9) is the evaluation of $mIoU_{\text{label}}$ on each class in the validation dataset. Fig. 9 is a pictorial representation of (9) in the case evaluation of water area $mIoU_{\text{water}}$

$$mIoU_{\text{label}} = \frac{1}{2} \times \left(\frac{TP_{\text{label}}}{TP_{\text{label}} + FP_{\text{label}} + FN_{\text{label}}} + \frac{TP_{\text{background}}}{TP_{\text{background}} + FP_{\text{background}} + FN_{\text{background}}} \right) \tag{9}$$

where

1) true positive (TP_{label}): The area of intersection between ground truth (GT_{label}) and segmentation mask (S_{label}) on each label (water, land, cloud). Mathematically, this is the logical AND operation between GT_{label} and S_{label};
2) false positive (FP_{label}): The predicted area outside GT_{label}. This is the logical OR of GT_{label} and S_{label} minus GT_{label};
3) false negative (FN_{label}): The number of pixels in the GT_{label} area that the model failed to predict. This is the logical OR of GT_{label} and S_{label} minus S_{label}.

Mean average precision (mAP_{label}) is the average over multiple $mIoU_{\text{label}}$, corresponding to the average precision $mIoU_{\text{label}}$ from 0.5 to 0.95 with a step size of 0.05. This section uses mAP_{water} to ignore the water bodies that differ completely between the recognized and the ground truth that focuses on determining the accuracy of the water body identification where $mIoU_{\text{water}} > 0.5$. In the comparison, the best deep learning model in water body recognition must have the highest $mIoU_{\text{water}}$ value, then the mAP_{water} value is considered.

Table III compares the recognition evaluations of the RGB and RGBNIR on U-NET. When adding NIR channels to the input tensor, mAP_{water} and $mIoU_{\text{water}}$ are increased by 2.7% and 0.01, respectively. Thus, NIR plays an important role in identifying bodies of water (rivers, lakes, and floods).

TABLE III
COMPARISON OF THE PREDICTION EVALUATION BETWEEN RGB AND RGBNIR ON THE ORIGINAL U-NET

	mAP_{land}	mAP_{water}	mAP_{cloud}	$mIoU_{land}$	$mIoU_{water}$	$mIoU_{cloud}$
RGBNIR	94.2%	**96.2%**	82.5%	0.916	**0.658**	0.762
RGB	94.4%	**93.5%**	84.8%	0.920	**0.648**	0.779

TABLE IV
COMPARISON OF WATER RECOGNITION OF THE PROPOSED LOSS WITH OTHERS

	U-NET[13] $\mathcal{L} = 4$	Toward onboard U-NET [2] $\mathcal{L} = 3$	Toward onboard U-NET [2] $\mathcal{L} = 3$
Loss function	BCE loss	combo [2]	***proposed***
$mIoU_{land}$	0.920	0.931	0.940
$mIoU_{water}$	***0.648***	***0.693***	***0.800***
$mIoU_{cloud}$	0.779	0.811	0.769
mAP_{land}	94.4%	94.8%	97.9%
mAP_{water}	93.5%	96.8%	87.4%
mAP_{cloud}	84.8%	88.0%	82.7%

Table IV compares the proposed loss function with other loss functions. The proposed loss function increases the accuracy of water recognition to 15.2% compared to the original U-NET [13], and 10.7% compared to the combo loss function used in [2].

This section uses two flood events representing the inference process based on satellite images, including Uganda [43] and Corail/Pestel Communes [44]. Figs. 10 and 11 show the results of surface water recognition. Figs. 10(a) and 11(a) are optical RGB images (captured by the Sentinel-2 satellite). Figs. 10(b) and 11(b) are ground truth segmentation maps from the World Flood dataset [2]. Figs. 10(c) and 11(c) are the recognition images that result from the proposed SWRNet model. The evaluation focuses on the accuracy of surface water recognition and the model's ability to effectively handle tiny/small water surface regions and differentiate river channels from water surfaces.

B. Flood Identification/Prediction

This section describes the method used for identifying flood areas and issuing warnings based on the surface water recognized by proposed SWRNet. Determining flood-prone areas relies on two categories: Permanent water and extended regions. Permanent water regions include consistent water bodies, such as lakes, rivers, and reservoirs, which maintain their presence over time, showing limited sensitivity to short-term weather changes. In contrast, extended water regions pertain to territories that do not usually experience water cover but become submerged during occurrences, such as heavy rain or rapid snowmelt.

The Global Surface Water dataset [45] consists of 504 titled images, each covering a 10°×10° geographic area. It takes up 12.6 GB of storage and is suitable for satellite platforms. The

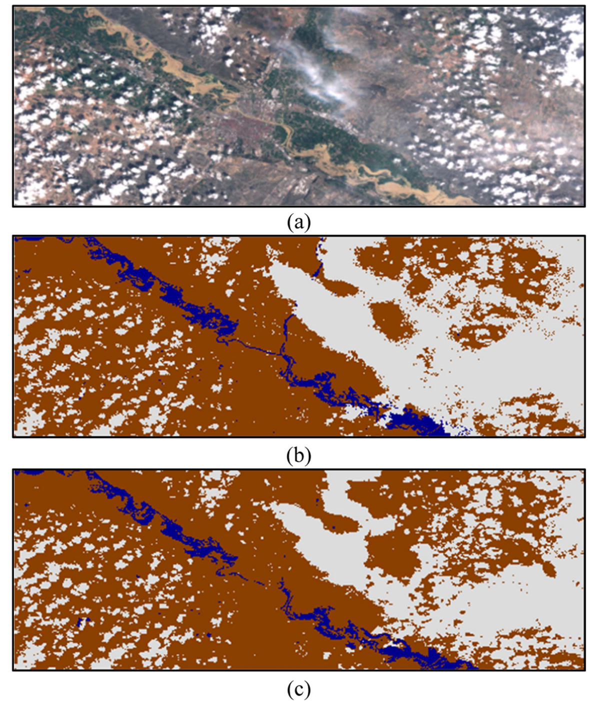


Fig. 10. Water/flood recognition in Uganda [43]. (a) Optical RGB image (Sentinel-2 satellite). (b) Ground truth. (c) Recognition image from the proposed SWRNet model.

dataset assists in identifying permanent water regions quickly using titled images. Figs. 12(a) and 13(a) represent the permanent water regions of Fig. 10 in Uganda [43] and Fig. 11 in Corail and Pestel Communes [44], respectively. The SWRNet model is used to identify surface water in real-time using satellite images captured by onboard satellites. By cross-referencing geographical coordinates of permanent water regions with identified surface features derived from satellite imagery by SWRNet, the goal is to assess the precise extent of the flooded area. The flood region is defined by a converse nonimplication of surface water recognized $P_{W_t}(x, y)$, Figs. 12(b), 13(b), and identifying permanent water regions – $\mathrm{W}(x, \mathrm{y})$, Fig. 12(a), Fig. 13(a) with the same geographical coordinates

$$\text{Flood} = \neg(W(x, \mathrm{y}) \leftarrow P_{W_t}(x, y)). \tag{10}$$

Equation (10) determines the flood area based on permanent and recognized surface water, which is the NOT $(W(x, \mathrm{y}))$ AND $P_{W_t}(x, y)$. This equation involves two binary operators, reflecting a design approach that prioritizes efficiency and compatibility with hardware resources. Consequently, flood areas are accurately identified onboard satellites, followed by the transmission of the corresponding flood map to the ground station. Fig. 12(c) and 13(c) are flood map. This strategy facilitates faster data transfer compared to the conventional method of transmitting optical image data.

Utilizing data gleaned from Figs. 10 to 15, the proposed model exhibits a capacity to discriminate the colors of diverse water bodies on Earth's surface. Significantly, it aptly identifies the characteristic yellow–brown tones indicative of delta river areas,

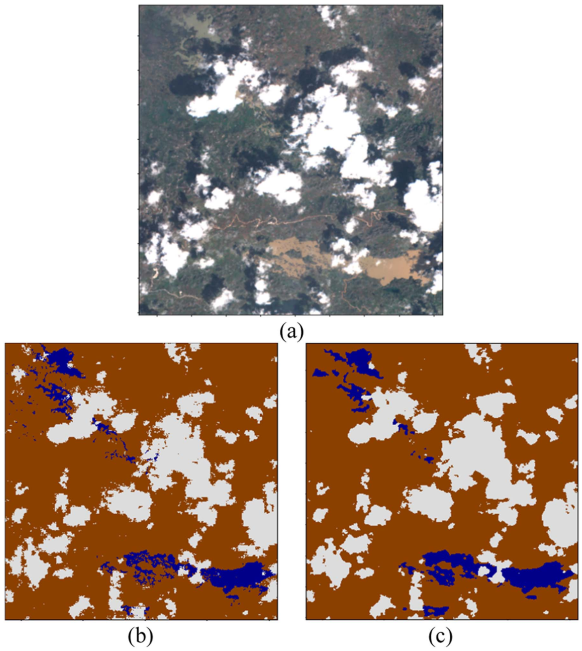


Fig. 11. Water/flood recognition in Corail and Pestel Communes [44]. (a) Optical RGB image (Sentinel-2 satellite). (b) Ground truth. (c) Recognition image from the proposed SWRNet model.

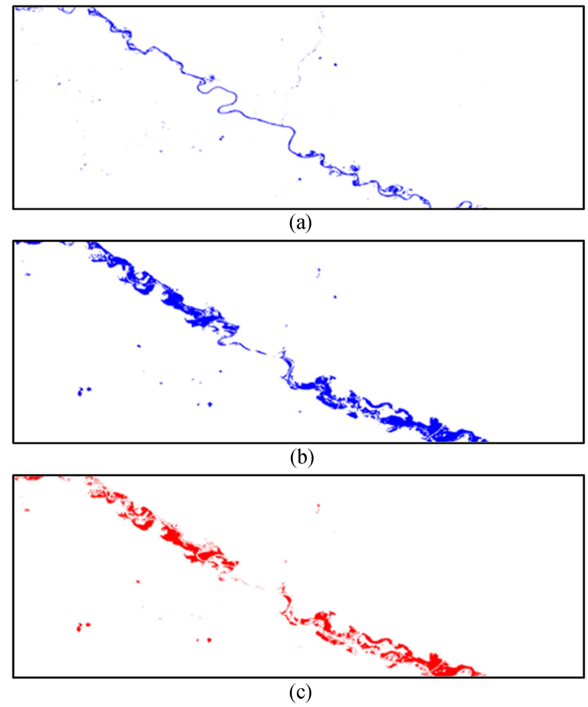


Fig. 12. Flood identification/prediction. (a) Permanent surface water. (b) Surface water area from Fig. 10(c). (c) Flood extent areas.

TABLE V
ACCURACY COMPARISON BETWEEN THREE DIFFERENT INPUT IMAGE SIZES

	mAP_{land}	mAP_{water}	mAP_{cloud}	$mIoU_{land}$	$mIoU_{water}$	$mIoU_{cloud}$
572 × 572	94.2%	**96.2%**	82.5%	0.916	**0.658**	0.762
256 × 256	94.4%	**96.3%**	86.0%	0.924	**0.685**	0.781
128 × 128	94.6%	**96.9%**	89.0%	0.930	**0.690**	0.817

as delineated in Figs. 10 and 11. This distinctive coloration is attributed to the suspended sediment, silt, and organic matter conveyed by the river. Concurrently, the model showcases its discernment of transparent blue hues associated with lakes, as exemplified in Figs. 14 and 15. This adeptness in color discrimination underscores the model's efficacy in discerning a spectrum of aquatic features across varied landscapes, accommodating the nuanced color divergences inherent in both delta river and lake environments.

C. Cutting-Edge Deep Learning Model Small Surface Water Area Recognition Onboard Satellite

The object recognition accuracy is increased when reducing the input image size. However, an important factor in the early prediction of natural disasters is the speed of the warning; earlier warnings help people make timely decisions to save lives and protect property. This section aims to evaluate and compare the computational complexity of the proposed SWRNet model with other state-of-the-art methods, considering the deployment of the proposed model onboard satellites, with the goal of reducing the time required for flood event identification and early warning.

Table V evaluates the impact of input image size on water identification speed using the proposed deep learning model. Reducing the size of subimages from 572 × 572 to 256 × 256 increases the number of subimages by four times, as does reducing the size from 256 × 256 to 128 × 128. In the inference process (determining the flood area), the image captured by satellite sensor is split into subimages. If the processing is single-threaded, the speed of the flood area prediction process is increased by four times for each subimage size reduction with the same image size. If multithreading is carried out, the memory and resource usage for the satellite are increased by four times for each reduction in subimages size, power consumption is also significantly increased. However, the $mIoU_{\text{water}}$ water coefficient of subimages 256 × 256 is 0.027 times higher than that of 572 × 572 and only 0.005 lower than 128 × 128. Therefore, this study proposes to use 256 × 256 subimages in building a cutting-edge deep learning model for water body recognition.

Table VI compares the proposed deep learning model focused on water/flood areas to the original U-NET [13] and flood mapping with a machine learning model [2] based on U-NET.

The proposed loss function increases the accuracy in water recognition by 15.2% compared to the original U-NET and 10.7% compared to the loss equation used in [2]. When combining the reduction of network structures and the proposed loss function, the computational complexity of the network and parameter size are reduced by 18.66 and 8.86 times, respectively; $mIoU_{\text{water}}$ is increased by 14.1%. With lower computational complexity and smaller parameter size, the SWRNet model is suitable for deploying cutting-edge deep learning models with low bandwidth limitations on cube satellites, such as IRIS-B.

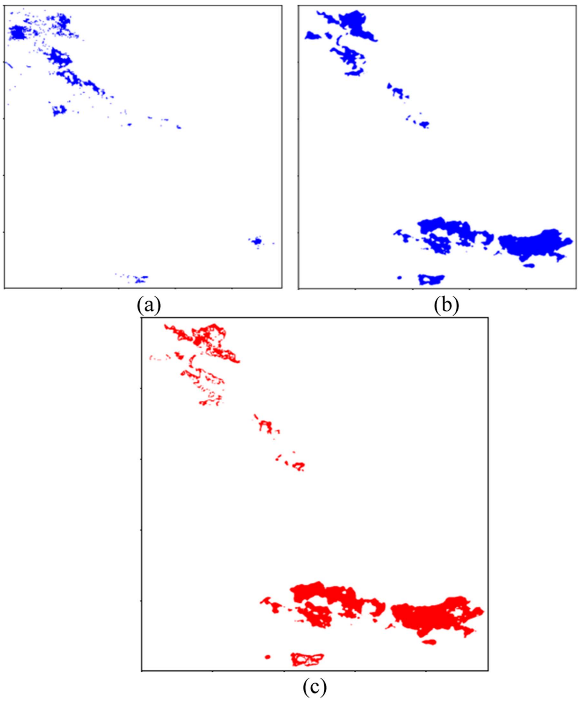


Fig. 13. Flood identification/prediction. (a) Permanent surface water. (b) Surface water area from Fig. 11(c). (c) Flood extent areas.

TABLE VI
COMPARISON OF DEEP LEARNING MODELS FOR FLOOD/WATER SEGMENTATION

	U-NET[13] $\mathcal{L}=4$	Toward onboard U-NET [2] $\mathcal{L}=3$	Toward onboard U-NET [13] $\mathcal{L}=3$	***Proposed* SWRNet**
Loss function	BCE loss	compound [2]	***proposed*** compound	***proposed*** compound
Computation complexity	199.31 GMac	42.51 GMac	42.51 GMac	***10.69*** GMac
Parameter size	17.27 M	7.78 M	7.78 M	***1.95*** M
$mIoU_{land}$	0.920	0.931	0.940	0.936
$mIoU_{water}$	***0.648***	***0.693***	***0.800***	***0.789***
$mIoU_{cloud}$	0.779	0.811	0.769	0.756
mAP_{land}	94.4%	94.8%	97.9%	97.2%
mAP_{water}	93.5%	96.8%	87.4%	89.9%
mAP_{cloud}	84.8%	88.0%	82.7%	82.4%

Tables VII and VIII compare proposed SWRNet with other state-of-the-art deep learning models.

Tables VI, VII, and VIII present a comparative analysis of various deep learning models utilized for flood/water segmentation in satellite imagery. The models evaluated in the study include DeepLab v3 ResNet 101 [7], SegNet [46], FCN ResNet 101 [9], original U-NET [13], simple U-NET [2], [13],

TABLE VII
COMPARISON OF DEEP LEARNING MODELS FOR FLOOD/WATER SEGMENTATION

	DeepLab [7]	SegNet [46]	FCN [9]	***Proposed* SWRNet**
Computation complexity	62.89 GMac	40.19 GMac	56.59 GMac	***10.69*** GMac
Parameter size	61.0 M	29.44 M	54.31 M	***1.95*** M
$mIoU_{land}$	0.869	0.853	0.864	0.936
$mIoU_{water}$	***0.001***	***0.415***	***0.001***	***0.789***
$mIoU_{cloud}$	0.782	0.689	0.756	0.756
mAP_{land}	97.9%	89.9%	97.2%	97.2%
mAP_{water}	0.1%	66.4%	0.1%	89.9%
mAP_{cloud}	84.8%	86.1%	88.9%	82.4%

TABLE VIII
COMPARISON OF UNET-BASED MODELS FOR FLOOD/WATER SEGMENTATION

	Attention-UNet [20][21]	Trans-UNet [22]	*MALU-Net* [26]	***Proposed* SWRNet**
Computation complexity	66.68 GMac	17.44 GMac	***92.0*** *MMac*	***10.69*** GMac
Parameter size	34.88 M	25.3 M	***178.03*** *k*	***1.95*** M
$mIoU_{land}$	0.853	0.854	0.865	0.936
$mIoU_{water}$	***0.461***	***0.528***	***0.522***	***0.789***
$mIoU_{cloud}$	0.334	0.303	0.639	0.756
mAP_{land}	89.3%	90.8%	91.9%	97.2%
mAP_{water}	95.0%	90.4%	79.7%	89.9%
mAP_{cloud}	52.6%	49.53%	74.5	82.4%

Attention-UNet [20], [21], Trans-UNet [22], MALUNet [26], and the proposed SWRNet. The evaluation is conducted based on multiple criteria, including computation complexity, parameter size, and accuracy, which are measured using the IoU metric. The findings reveal that MALUNet exhibits lower computation complexity compared to the proposed SWRNet. However, the accuracy of MALUNet is comparatively lower as well. In contrast, the proposed SWRNet demonstrates superior performance, showcasing lower computation complexity of 10.69 GMac and smaller parameter size of 1.95 M in comparison to the other evaluated models. Notably, the proposed SWRNet also achieves the highest accuracy (IoU) of 0.789 in the context of water segmentation, underscoring its effectiveness in accurately identifying flooded areas in satellite imagery.

These results highlight the promising performance of the proposed SWRNet model in achieving a favorable balance between computation complexity, parameter size, and accuracy

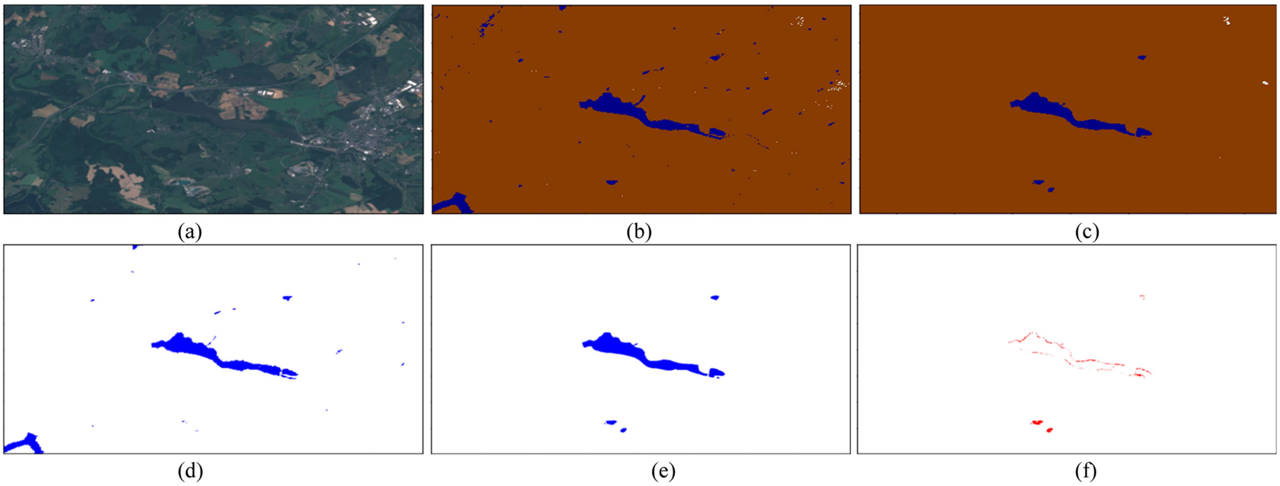


Fig. 14. Surface water recognition - test case 3. (a) Optical RGB image (by Sentinel-2 satellite). (b) Ground truth. (c) Recognition image by the proposed SWRNet model. (d) Permanent surface water from Global Surface Water dataset [45]. (e) Water areas from (c). (f) Flood areas.

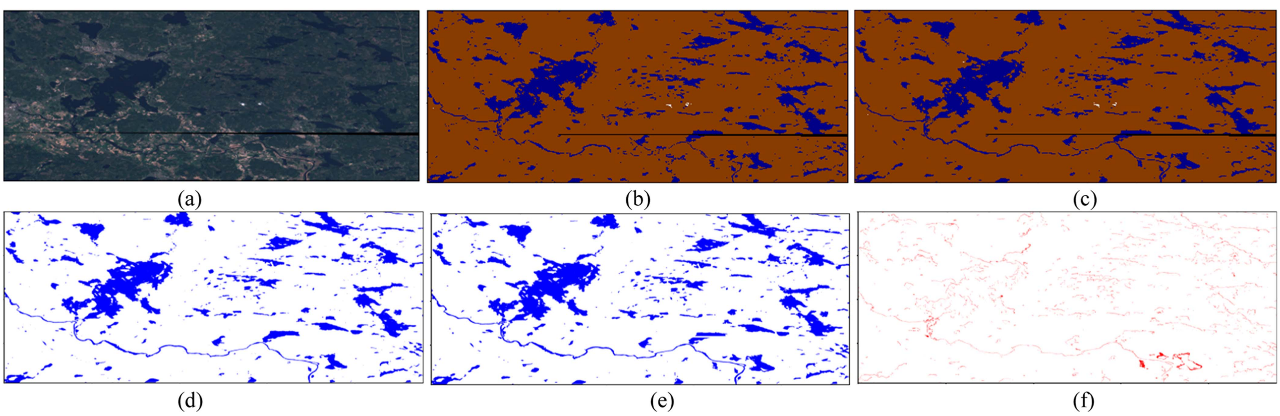


Fig. 15. Surface water recognition - test case 4. (a) Optical RGB image (by Sentinel-2 satellite). (b) Ground truth. (c) Recognition image by proposed the SWRNet model. (d) Permanent surface water from Global Surface Water dataset [45]. (e) Water areas from (c). (f) Flood areas.

for flood/water segmentation tasks in the context of satellite imagery. The findings suggest that the proposed SWRNet model holds potential for practical implementation onboard satellites, making it a promising solution for accurate flood/water segmentation in satellite imagery.

IV. Discussion

This article proposes a cutting-edge model to recognize surface water areas directly on satellite payloads. The computational complexity of the proposed model is 18 times lower, whereas the accuracy of water detection is increased by 14.1% compared to the original U-NET model. This makes the identified speed 18 times faster with more accuracy than other models. The appropriate proposed method is edge-computing on the satellite payload.

The advantage of deploying a cutting-edge deep learning model on satellite payload is that the results can be produced quickly, reducing the download data. However, the main limitation is that the onboard deep learning models are difficult to change, as done on raw image data at the ground station. To solve this problem, our CubeSAT builds an upload band to update deep learning model's weights and labels. Although the upload band is small, it still meets the deep learning model updating in 5–7 days.

Utilizing drones for water body identification offers several advantages, including lower operating costs, ease of operation, and customization when compared to direct satellite-based approaches. However, for large-scale studies or extensive water bodies, the need for multiple flights can result in heightened time and expenses. Notably, in the context of predicting weather-related phenomena, drones are vulnerable to adverse weather conditions, such as wind, rain, and fog, which can curtail their capacity to conduct flights and gather data effectively. Consequently, employing satellite images and implementing water detection directly from satellite payloads present numerous benefits in contrast to drone-based methodologies.

V. Conclusion

This article not only proposes a cutting-edge deep learning model on satellite payload focused on reducing the time of water recognition, adding NIR channel to RGB increases the accuracy of surface water recognition, but the proposed loss function also focuses on the identification of small objects (small areas of surface water in satellite images). The accuracy of recognition is significantly improved, moreover the complexity of the deep learning model is reduced many times. The proposed model effectively discerns the colors of diverse water bodies, distinguishing characteristic yellow–brown tones in delta river areas and transparent blue hues in lakes, showcasing its suitability for varied aquatic features across different landscapes. Deploying the cutting-edge model on satellite payload reduces transmission data from satellites to ground stations.

In conclusion, the limitations of delays in data transmission for remote-sensing-based surface water recognition can be overcome by employing cutting-edge deep learning models that are suitable for onboard satellite processing. The proposed SWRNet model offers a promising solution for real-time identification with reduced computational complexity and improved speed, leading to more timely and effective early warning systems.

In addition to simplifying the computational complexity and increasing the accuracy of the cutting-edge deep learning model, implementing algorithms and optimizing computing devices reduce the power consumption of the satellite payload. CPUs and GPUs consume huge amounts of power and are unsuitable for small satellites. Thus, this research focuses on implementations with field-programmable gate arrays and computing-in-memory hardware technologies, which speed up cutting-edge deep learning models with low power consumption (limited battery) of satellite payloads.

More experiment results of surface water recognition and flood identification/prediction from satellite images.

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Trong-An Bui (Member, IEEE) received the Ph.D. degree in electrical engineering from the Electrical Engineering Department, National Chi Nan University, Nantou, Taiwan, in 2022.

He is currently a Postdoctoral Researcher with the Department of Electronic and Computer Engineering, National Taiwan University of Science and Technology, Taipei, Taiwan. His current research interests include neural networks and artificial intelligence, computer vision and image processing, video coding and data compression, remote sensing, and satellite imagery.

Pei-Jun Lee (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the Department of Electrical Engineering, National Taiwan University, Taipei, Taiwan, in 2004.

In 2015, she was promoted to a Distinguished Professor with the Department of Electrical Engineering, National Chi Nan University, Nantou, Taiwan. Since August 2021, she has been a Professor with the Department of Electronics and Computer Engineering, National Taiwan University of Science and Technology, Taipei, Taiwan. Her research interests include 2-D/3-D image conversion, 3-D/multiview video coding, video communication, video compression, image processing, robot fish, and FPGA system application.

Dr. Lee was the recipient of the First Runner-Up and the Outstanding Advisor of the Macronix Golden Silicon Awards Semiconductor Design and Application Competition from MXIC Company Ltd., in 2008.